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POWER FLOW STUDIES FOR RADIAL DISTRIBUTION SYSTEMS WITH DISTRIBUTED GENERATION

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ABSTRACT

This paper presents power flow solution techniques and analyses of radial distribution systems with distributed generation incorporated in their network. The result of the analysis gives the power system static states in terms of bus voltage magnitudes and voltage angles, active and reactive powers as well as system losses. Since conventional power flow methods have proven to be inefficient for radial distribution systems, a more robust algorithm for systems with distributed generation integrated in the network is proposed which is achieved by developing power flow program and graphical user interface (GUI) module in Matlab. The effectiveness of the proposed approach is tested on IEEE 33-bus system and results are presented in a graphical user interface form for better visualization.

KEYWORDS: Power flow, Load flow, Distributed Generation, Newton-Raphson, fast decoupled, Guess-seidal.

1. INTRODUCTION

Power flow sometimes referred to as load flow is a solution of static operating condition equations of power system under various load demands, source supply and network configuration, provided that the load consumption or demand is known at all buses. The analysis involves determination of the voltage magnitude and phase angle at each bus, power flow in each branch for transformers and lines, power consumption from each power generation source, active power or reactive volt-amperes generation or consumption of each shunt element and system losses. The results of the analysis if determined from the system model can be adjusted to ensure that all loads are supplied with quality power that falls within specified operational limit of the system. Adjustments can be on the following for optimum operational specifications based on generator set points, transformer tap changer adjustment, reactive power compensation or Making emergency power available to satisfy voltage and frequency operating limits. Also, the power flow solutions are useful for checking of under/over voltage, overloads and assessment of losses in a network. It is also important for future expansion planning, system stability studies and in determining the optimal operation of existing system especially with appearance of distributed generation (DG) in distribution systems. Its advantage can also be seen when making settings for protection devices that can ensure system security. It is obvious that in the near future most distribution systems would be networked or looped with more points of power injection in the form Distributed Generation resources as reported by (Heydt, 2010).

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Hence in future Distributed Power Flow (DPF) is required to solve the three phase unbalanced load flow for both meshed and radial operation scenarios of the distribution network integrated with distributed and active resources such as renewable power generations, loads, storages and electric vehicles. This creates interest on the review of research developments in the power flow studies on the distribution networks as reported by (Balamurugan and Srinivasan, 2011).

2. POWER FLOW SOLUTION TECHNIQUES

An overview of different techniques employed for power flow study under various operating condition are briefly considered in this work. Due to the nonlinearity and the difficulty involved in the analysis of power flow equations various methods are available and the most widely adopted among them are as shown in figure 1. The speed of convergence of these methods is of extreme importance as well as the cost of computer time especially when employed in schemes for automatic control of power systems (Ranjan,. et al, 2004).

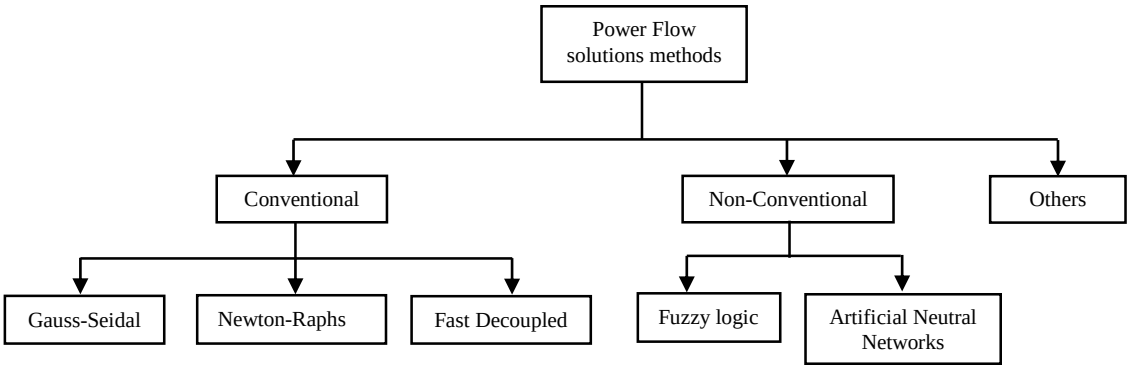


Figure 1: Power Slow Solution Method

As indicated in the figure 1, the power flow solutions methods are categorized into conventional, non-conventional and others. The others, are any other methods employed in power flow solution that does not fall under any of the two categories mentioned which among them are the altered or modified Newton's methods.

2.1 Conventional Methods

The conventional methods are the Gauss-Seidel (GS), Newton Raphson (NR) and Fast Decoupled (FD) Load Flow methods. The earliest computational load flow method was the Gauss-Seidel method however, it suffered from poor convergence problem and the number of iterations required before convergence is normally very high. The rate of convergence can be influenced by the use of acceleration factor for making convergence speed to increase.

It was around 1960 that a better method that can assure convergence was introduced known as Newton Raphson (NR) method. This method can converge after few iterations for typical networks unlike Gauss-Seidal method. The NR method started losing its fast convergence ability and more studies were focused on better methods that can promise faster convergence. It was around the same time that Stott and Alsac (Garcia and Zago, 1996) introduced Fast Decoupled method which

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improved computational speed for large power systems load flow solutions. By considering the decoupling characteristic between active power with voltage, and reactive power with angle. It is well-known for its fast convergence characteristic and requires much less memory storage. When the decoupling principle did not hold, the NR method was the preference as reported by (Brown and Tinney 1957 and Srinivasa Rao et al. 1998).

However, when these methods are applied to radial distribution systems or weakly meshed systems with their high R/X ratio slow convergence or divergence are experienced and these affected the stability of the methods (Stagg and El-Abiad 1968, and Vander et al 2007). Hence, the possibility of having no solution increases for such system and this led to the development of alternative methodologies, based on the NR iterative scheme (Costa, et al 2001).

2.2 Non-Conventional Methods

Conventional methods were used to effectively solve power flow problems. But with the recent introduction of flexible alternating current transmission system (FACTS) and other devices and deregulation of a power sector, the conventional concepts can no longer be adequate enough. So Power Flow solution have become complex and Artificial Intelligence (AI) methods have emerged which can solve highly complex problems by making use of intelligent computer based programs.

One of such method is Fuzzy Logic method (FL) that is based on logic that can represent the proposition with degrees of truthfulness and false hold. It is a very powerful tool for the solution of nonlinear algebraic equations. A fuzzy logic based power flow method has been proposed by (Shishir, et al 2006) in which voltage magnitudes and angles at different buses of power systems are fast computed. A similar work on fuzzy power flow solution was done by considering reactive power limit violations at voltage controlled buses, uncertainties in voltage dependent load models, load forecast and system parameters (Bijwe, et al, 2005). Another superior speed method is the Artificial Neural Network power flow solution which is based on the concept of neural network that serves as a powerful tool for solving nonlinear power flow equations was proposed by (Arunagiri, et al, 2006 and Paucar, et al, 2006) for the evaluation of voltage magnitudes and angles for a power flow problem.

2.3 Other Power Flow Solution Methods

A number of other power flow solution methods have been successfully employed apart from those methods mentioned above. Survey has shown that faster power flow solution is obtained with other miscellaneous method. For example (Sasson et al, 1971) have altered Newton's method to obtain a faster convergence than with the normal approach. This was achieved by comparing the method with the non-linear programming approach for the solution of load flow problem in terms of convergence process and this feature helps the method to converge for cases when the normal Newton's method diverges.

3. POWER FLOW WITH DISTRIBUTED GENERATION

Integration of DG causes bi-directional power flow which reduces the capacity of feeder and transmission line. On the other hand the benefits associated with distributed generation include the reduction of power loss, better voltage support, peak shaving and the improvement of overall efficiency, stability and reliability (Sambo et al 2003, Agbetuyi et al 2013). The distribution system is slightly different from transmission system in the sense that it is radial or weakly meshed and the lines have high R/X ratio and it is usually unbalance by nature. Many algorithms on power flow for distribution system with high penetration of DG have been proposed by many researchers. Based on researchers' findings, it can be categorized into node based and branch based methods (Wu and Zhang 2008).

In the node based methods, node voltage or current injections are employed as state variables to solve the power flow problems (Garcia, and Zago, 1996), (Garcia, and Pereira, 2000), (Teng and Chang, 2002). This includes network equivalence method, Z-bus method, Newton-Raphson algorithm and fast decouple algorithm. On the other hand, the branch based involves the use of branch currents or powers as the state variable for the power flow solution (Thukaram, et al 2000, Bhutad, et al 2003, Ranjan, et al. 2004). The technique is applicable to sweep based and loop impedance methods.

With the continuous penetration of distribution generation (DG) into the distribution network a lot of research works have been done with the intention of investigating the impact of DG on power quality and its control capabilities to provide fast voltage regulation. It was shown that with proper control method, inverter based DG will give better and fast load voltage regulation as well as mitigation of unbalanced voltage disturbance (Mohamed, and El-Saadany 2009).

3.1 Integrating DG Units in Power Flow Solution

Since the DG units are modeled as either P and Q control bus (PQ) or P and V control bus (PV) their incorporation into power flow will involve use of active and reactive powers as flow variables instead of complex currents as variables. In the case of PV model, the reactive power limits of the generator are constantly checked so that it is within limits otherwise the model will become a PQ model. The root bus is considered as the slack bus with known voltage magnitude and angle. It must also be assumed that the initial voltages for all the other nodes are equal to the root node voltage and the initial power loss of all the branches is equal to zero.

For the PQ model the DG is treated as negative load in the power flow problem. However, for PV model some additional process is required as the generator terminal voltage is controlled by the specification of positive sequence components for example. So for a PV bus, the three-phase active power output and positive sequence voltage of the generator must be specified. This technique was proposed by (Baran and Wu 1989) for optimal capacitor placement and sizing in radial distribution system and the same was employed for the placement and sizing of DGs with n bus, laterals and sub-laterals as shown in Figure 2 for radial distribution system.

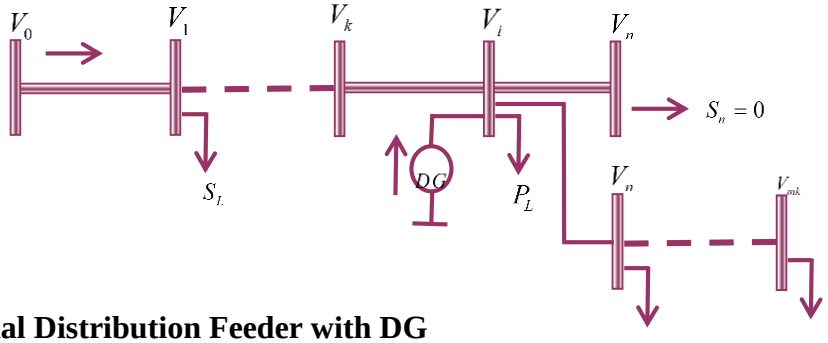


Figure 2: Radial Distribution Feeder with DG

The load flow is performed by making the following assumptions;

1. The distribution lines one modeled as series impedance given by:

$$Z_i = R_i + jX_i$$

Where; R_i and X_i are the resistance and reactance of the line i respectively.

2. The DG is modeled as PQ with a constrained power and referred to as Negative load or as PV model and the voltage magnitude is limited to a specified value. The generation at bus i is modeled as constant power source given by;

$$S_{Gi} = P_{Gi} + jQ_{Gi} \quad (1)$$

Where; P_{Gi} and Q_{Gi} are the active and reactive powers of the generation at bus i respectively.

The Power supplied from the substation is primarily specified as:

$$S_O = P_o + jQ_O \quad (2)$$

Where; P_o and jQ_O are the active and reactive power supplied from the substation respectively.

Then the active and reactive power as well as the voltage at the receiving end of the first branch can be express as follows:

$$S_1 = S_o - S_{loss1} - Z_1 \frac{P_O^2 + Q_O^2}{|V_O|^2} - S_{L1} \quad (3)$$

$$V_1 \angle \theta_O = V_O - Z_1 \frac{S_O^*}{|V|} \quad (4)$$

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Where; S_{loss1} and Z_1 are power loss and impedance for bus 1 respectively.

The voltage at the receiving end can further be expressed as:

$$V_1 = V_O - \frac{P_O - jQ_O}{{V_O}^2}(R_i + jX_i) = \left(V_O - \frac{P_O R_i + Q_O X_i}{V_O}\right) - j\left(\frac{P_O X_i - Q_O R_i}{V_O}\right) \quad (5)$$

$$\theta_i = -\tan^{-1}\left({}^{a_1}/_{a_2}\right) \quad (6)$$

Where $a_1 = \frac{P_O X_i - Q_O R_i}{V_O}$, and $a_2 = \frac{P_O R_i + Q_O X_i}{V_O}$

The active (P) and reactive (Q) power flowing can similarly be obtained at the receiving end branch $(i+1)$, as P_{i+1} and Q_{i+1} and the voltage magnitude at the sending end $|V_{i+1}|$ for the nodes can be expressed as;

$$P_{i+1} = P_i - R_{i+1}\frac{P_i^2 + Q_i^2}{|V_i|^2} - P_{Li+1} + P_{Gi+1} \quad (7)$$

$$Q_{i+1} = Q_i - X_{i+1}\frac{P_i^2 + Q_i^2}{|V_i|^2} - Q_{Li+1} + Q_{Gi+1} \quad (8)$$

$$|V_{i+1}|^2 = |V_i|^2 - 2(P_{i+1}.P_i + X_{i+1}.G_i) + (R_{i+1}^2 + X_{i+1}^2)\frac{P_i^2 + Q_i^2}{|V_i|^2} \quad (9)$$

At the end of the main feeder $P_n = 0\ and\ Q_n = 0$

Where $P_i Q_i$ are active and reactive powers following into the receiving bus i + 1

$|V_i|$ is the bus voltage magnitude at node i.

Similar formulation for the laterals can be achieved by using the line flow equations and the new terminal conditions as:

$$V_{ko} = V_{k,}\ P_{kn} = 0\ and\ Q_{kn} = 0$$

An iterative procedure is employed that will first require inputting the distribution system data that includes the voltage, number of nodes, segments and load connected to each node. Based upon the information, initial settings for power flow calculations such as specifying the supplied power to the substation and others. The calculation involves the calculation of the real and reactive power at each node for the main substation feeder and lateral feeders as well as voltage magnitudes and angles for all the nodes. If the results obtained are within the specified limits, then the power flow, line flow and loss for each lateral and main feeder nodes are displayed.

If on the other hand the results obtained are outside the specified limits, then the difference between the kth node real and reactive power and nth node of the sending end lateral is evaluated. Similar evaluation is also conducted for all nodes up to the last node of the main feeder and the process of power flow calculations is repeated as shown in Figure 3 for obtaining unique solution for the node voltage and line active and reactive power flow using Newton – Raphson method.

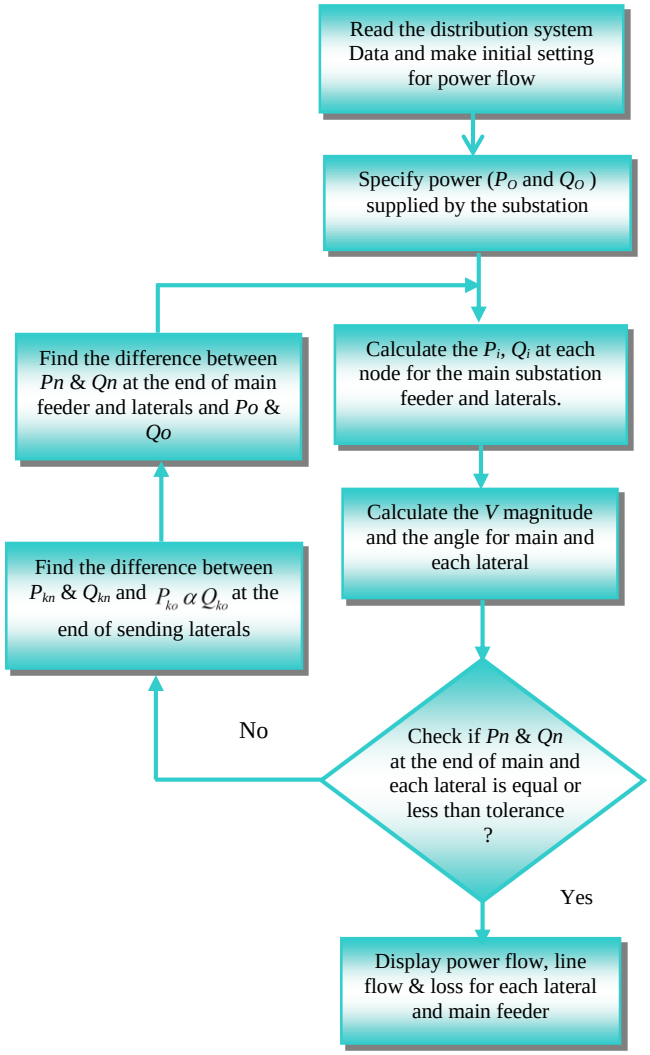


Fig 3.0: Power Flow Solutions with DG Algorithm

4.0 RESULTS AND DISCUSSIONS

The proposed tool employed for the simulation of the power flow studies is the Power Flow Analysis and distributed generation Optimization for radial distribution networks (PFADGOTRDN) purposely developed for power flow analysis and Distributed Generation

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optimization for radial distribution system in MATLAB environment using graphical user interface as shown in figure 4.

The package PFADGOTRDN consists of power flow solutions and DG optimization window which can allow system planner not only to obtain power flow solutions, but also finds optimal locations and sizes for DGs in radial distribution network.



Figure 4: Main Window of the PFADGOTRDN Software

The program routines are all in MATLAB and the main window is as shown figure 5 with three divisions into system activation section, power flow analysis section and DG optimization section.

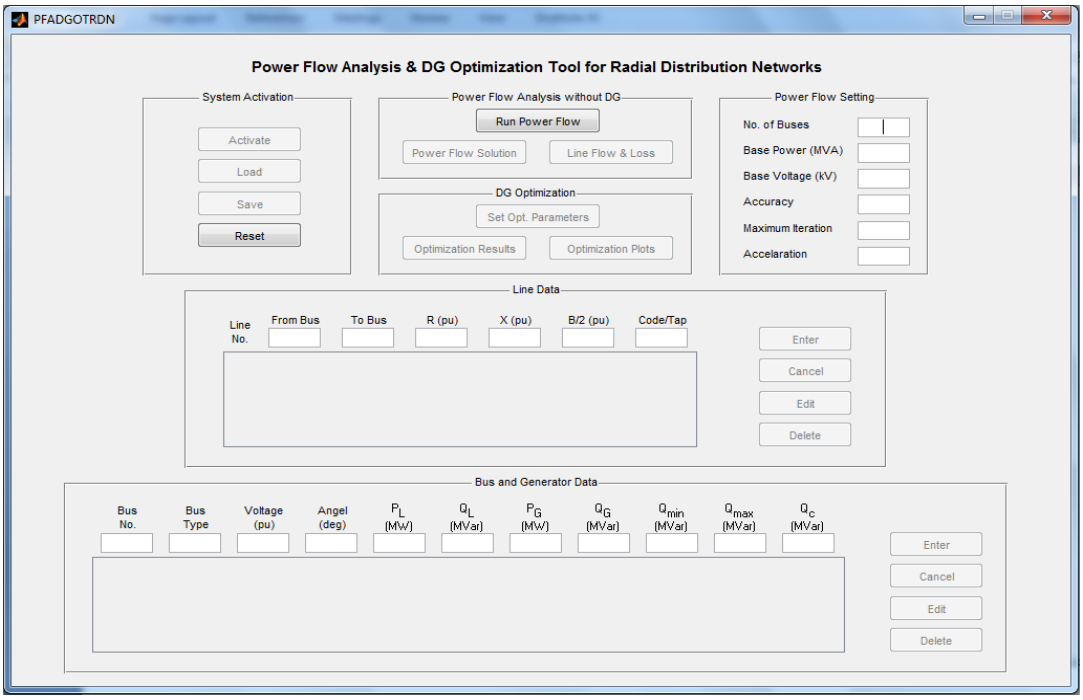


Figure 5: Manual Data Entry Mode

For the purpose of this software validation, a **33-bus IEEE test System** data are used for the running of the power flow solution and the system loaded with the system data is as shown in figure 6.

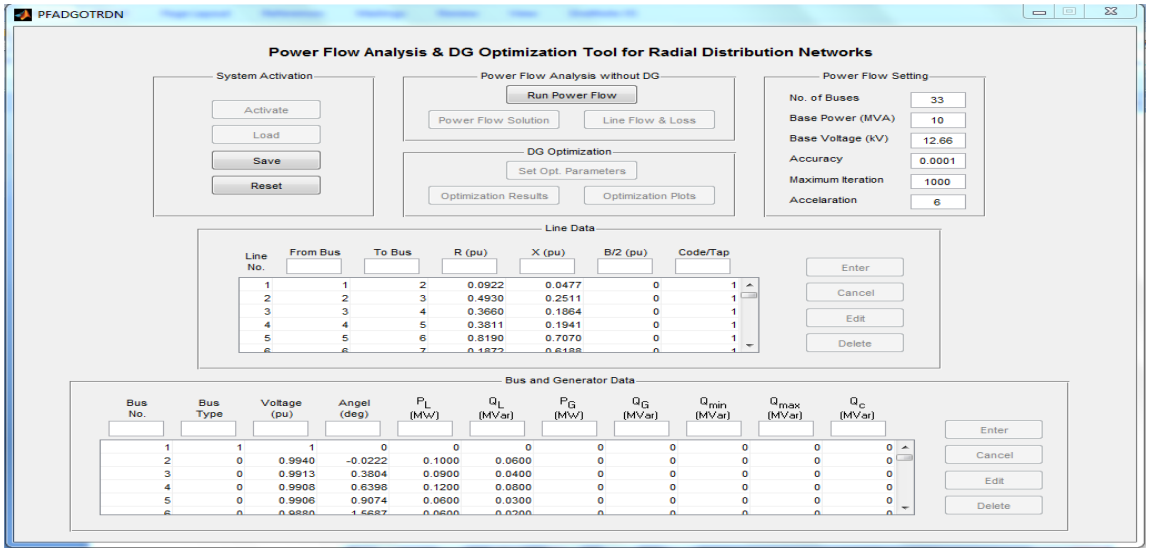


Figure 6: PFADGOTRDN loaded with 33-bus system

On clicking the power flow solution button the result for the solution is displayed as shown in figure 7 with the values for the bus voltages, load demand and generation power. It can be observed that the package gives the total system load to be 3.715 MW, while the generation turns out to be 3.9182 MW.

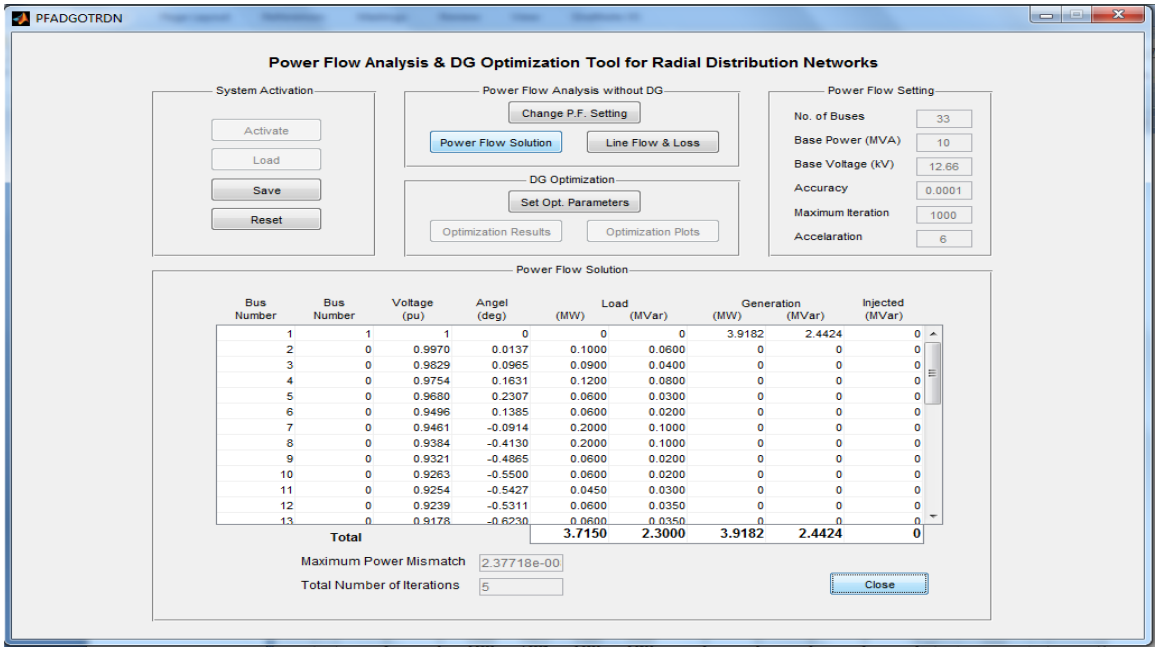


Figure 7: Power flow Solution Results for 33-bus system

The power loss result is also displayed by clicking the line flow and loss button as shown in figure 8.

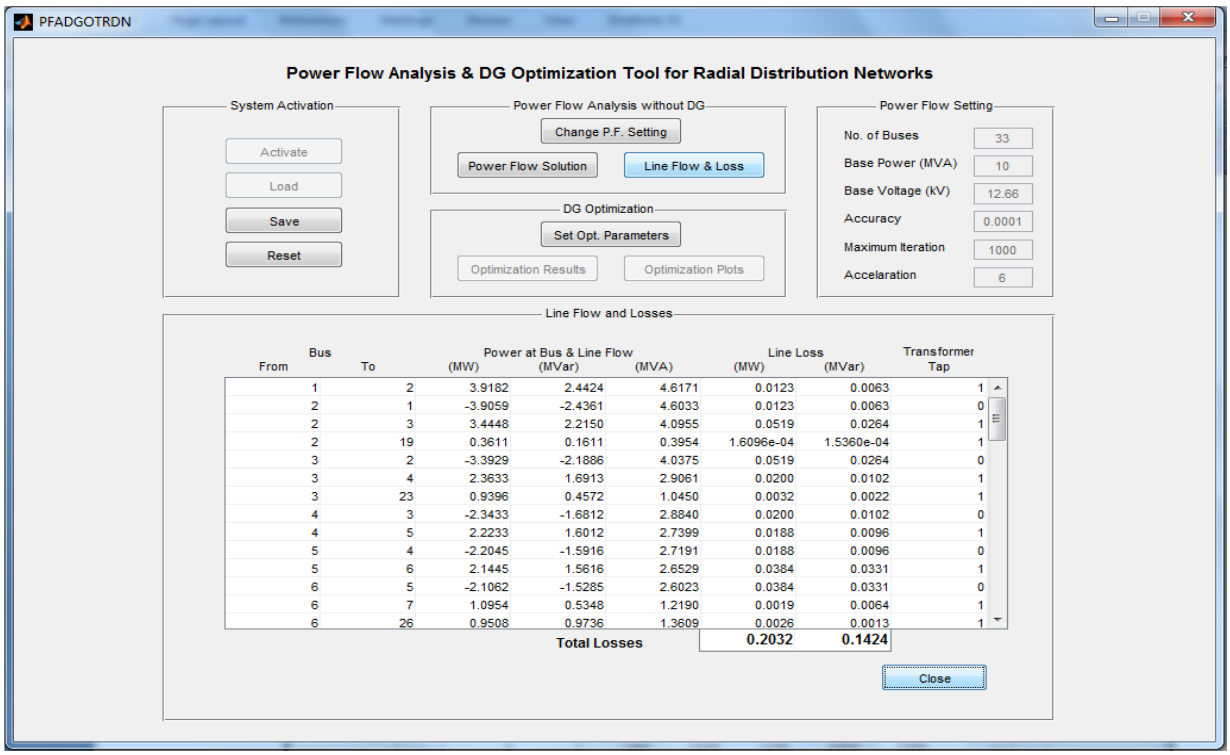


Figure 8: Line flow and losses results for 33-bus system

From the result the power at each bus and line losses are presented and the total real power loss is 0.2032 MW, while the total reactive loss is 0.1424 MVar.

5. CONCLUSION

This paper has tried to review the research developments in distribution system power flow with DG model integrated as source of power. The power flow methods were first classified as it applies to distribution system power flow solution with DG integrated. Secondly an algorithm was developed for power flow analysis with DG and without DG inform of simulation software for power flow studies called Power Flow Analysis and Distributed Generation Optimization for Radial Distribution Networks (PFADGOTRDN). The algorithm is found to be efficient practically and convergence is guaranteed. The effectiveness of the proposed software was verified using IEEE 33-bus system and results presented in a graphical user interface form for visualization.

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